



Universidad de Costa Rica

Escuela de Ciencias de la Computacion e Informatica

CI-0123 Proyecto Integrador de Sistemas Operativos y Redes de Comunicación
de Comunicación de datos

Conexión L2 con L3

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1.Configuracion del L3

```
!  
version 12.2  
no service pad  
service timestamps debug uptime  
service timestamps log uptime  
no service password-encryption  
!  
hostname L3-0403  
!  
!  
no aaa new-model  
ip subnet-zero  
ip routing  
!  
!  
!  
!  
no file verify auto  
spanning-tree mode pvst  
spanning-tree extend system-id  
!  
vlan internal allocation policy ascending  
!  
interface FastEthernet0/1  
  switchport trunk encapsulation dot1q  
  switchport trunk allowed vlan 230,240  
  switchport mode trunk  
!  
interface FastEthernet0/2  
  switchport trunk encapsulation dot1q  
  switchport trunk allowed vlan 230,240  
  switchport mode trunk  
!  
interface FastEthernet0/3  
!  
interface FastEthernet0/4
```

!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!

```
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
  no ip address
!
interface Vlan230
  ip address 172.16.123.62 255.255.255.240
!
interface Vlan240
  ip address 172.16.123.78 255.255.255.240
!
ip classless
ip http server
!
!
control-plane
!
!
line con 0
line vty 5 15
!
!
end
```

2.Configuration L2 ISLA 4

Building configuration...

Current configuration : 3288 bytes

```
!
version 12.2
no service pad
```

```
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname Pris
!
boot-start-marker
boot-end-marker
!
!
!
!
no aaa new-model
system mtu routing 1500
ip dhcp excluded-address 172.16.123.65 172.16.123.66
!
ip dhcp pool ISLA4
    network 172.16.123.64 255.255.255.240
    default-router 172.16.123.65
!
!
no ip domain-lookup
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
vlan internal allocation policy ascending
!
!
!
interface FastEthernet0/1
    switchport access vlan 240
```

```
switchport mode access
spanning-tree portfast
!
interface FastEthernet0/2
switchport access vlan 240
switchport mode access
spanning-tree portfast
!
interface FastEthernet0/3
switchport access vlan 240
switchport mode access
spanning-tree portfast
!
interface FastEthernet0/4
switchport access vlan 240
switchport mode access
spanning-tree portfast
!
interface FastEthernet0/5
switchport access vlan 240
switchport mode access
spanning-tree portfast
!
interface FastEthernet0/6
switchport access vlan 240
switchport mode access
spanning-tree portfast
!
interface FastEthernet0/7
switchport access vlan 240
switchport mode access
spanning-tree portfast
!
interface FastEthernet0/8
switchport access vlan 240
switchport mode access
spanning-tree portfast
```

!
interface FastEthernet0/9
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast

!
interface FastEthernet0/10
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast

!
interface FastEthernet0/11
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast

!
interface FastEthernet0/12
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast

!
interface FastEthernet0/13
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast

!
interface FastEthernet0/14
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast

!
interface FastEthernet0/15
 switchport access vlan 240
 switchport mode access
 spanning-tree portfast

!
interface FastEthernet0/16

```
switchport access vlan 240
switchport mode access
spanning-tree portfast
!
interface FastEthernet0/17
switchport access vlan 240
switchport mode access
spanning-tree portfast
!
interface FastEthernet0/18
switchport access vlan 240
switchport mode access
spanning-tree portfast
!
interface FastEthernet0/19
switchport access vlan 240
switchport mode access
spanning-tree portfast
!
interface FastEthernet0/20
switchport access vlan 240
switchport mode access
spanning-tree portfast
!
interface FastEthernet0/21
switchport access vlan 240
switchport mode access
spanning-tree portfast
!
interface FastEthernet0/22
switchport access vlan 240
switchport mode access
spanning-tree portfast
!
interface FastEthernet0/23
switchport access vlan 240
switchport mode access
```

```
spanning-tree portfast
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
switchport trunk allowed vlan 240
switchport mode trunk
!
interface GigabitEthernet0/2
!
interface Vlan1
no ip address
!
interface Vlan240
ip address 172.16.123.66 255.255.255.240
!
ip default-gateway 172.16.123.65
ip http server
ip http secure-server
vstack
!
line con 0
line vty 5 15
!
end
```

3.Configuración L2 ISLA 3

```
!
version 12.2
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname ISLA3
!
boot-start-marker
```

```
boot-end-marker
!
!
!
!
no aaa new-model
system mtu routing 1500
ip dhcp excluded-address 172.16.123.49 172.16.123.54
!
ip dhcp pool ISLA3
    network 172.16.123.48 255.255.255.240
--More--      default-router 172.16.123.49
--More--      !
--More--      !
--More--      no ip domain-lookup
--More--      !
--More--      !
--More--      !
--More--      !
--More--      !
--More--      !
--More--      !
--More--      spanning-tree mode pvst
--More--      spanning-tree extend system-id
--More--      !
--More--      vlan internal allocation policy ascending
--More--      !
--More--      !
--More--      !
--More--      interface FastEthernet0/1
--More--      switchport access vlan 230
--More--      switchport mode access
--More--      spanning-tree portfast
--More--      !
--More--      interface FastEthernet0/2
--More--      switchport access vlan 230
--More--      switchport mode access
--More--      spanning-tree portfast
```

```
--More--      !
--More--      interface FastEthernet0/3
--More--      switchport access vlan 230
--More--      switchport mode access
--More--      spanning-tree portfast
--More--      !
--More--      interface FastEthernet0/4
--More--      switchport access vlan 230
--More--      switchport mode access
--More--      spanning-tree portfast
--More--      !
--More--      interface FastEthernet0/5
--More--      switchport access vlan 230
--More--      switchport mode access
--More--      spanning-tree portfast
--More--      !
--More--      interface FastEthernet0/6
--More--      switchport access vlan 230
--More--      switchport mode access
--More--      spanning-tree portfast
--More--      !
--More--      interface FastEthernet0/7
--More--      switchport access vlan 230
--More--      switchport mode access
--More--      spanning-tree portfast
--More--      !
--More--      interface FastEthernet0/8
--More--      switchport access vlan 230
--More--      switchport mode access
--More--      spanning-tree portfast
--More--      !
--More--      interface FastEthernet0/9
--More--      switchport access vlan 230
--More--      switchport mode access
--More--      spanning-tree portfast
--More--      !
--More--      interface FastEthernet0/10
```

```
--More--      switchport access vlan 230
--More--      switchport mode access
--More--      spanning-tree portfast
--More--      !
--More--      interface FastEthernet0/11
--More--      switchport access vlan 230
--More--      switchport mode access
--More--      spanning-tree portfast
--More--      !
--More--      interface FastEthernet0/12
--More--      switchport access vlan 230
--More--      switchport mode access
--More--      spanning-tree portfast
--More--      !
--More--      interface FastEthernet0/13
--More--      switchport access vlan 230
--More--      switchport mode access
--More--      spanning-tree portfast
--More--      !
--More--      interface FastEthernet0/14
--More--      switchport access vlan 230
--More--      switchport mode access
--More--      spanning-tree portfast
--More--      !
--More--      interface FastEthernet0/15
--More--      switchport access vlan 230
--More--      switchport mode access
--More--      spanning-tree portfast
--More--      !
--More--      interface FastEthernet0/16
--More--      switchport access vlan 230
--More--      switchport mode access
--More--      spanning-tree portfast
--More--      !
--More--      interface FastEthernet0/17
--More--      switchport access vlan 230
--More--      switchport mode access
```

```
--More--      spanning-tree portfast
--More--      !
--More--      interface FastEthernet0/18
--More--      switchport access vlan 230
--More--      switchport mode access
--More--      spanning-tree portfast
--More--      !
--More--      interface FastEthernet0/19
--More--      switchport access vlan 230
--More--      switchport mode access
--More--      spanning-tree portfast
--More--      !
--More--      interface FastEthernet0/20
--More--      switchport access vlan 230
--More--      switchport mode access
--More--      spanning-tree portfast
--More--      !
--More--      interface FastEthernet0/21
--More--      switchport access vlan 230
--More--      switchport mode access
--More--      spanning-tree portfast
--More--      !
--More--      interface FastEthernet0/22
--More--      switchport access vlan 230
--More--      switchport mode access
--More--      spanning-tree portfast
--More--      !
--More--      interface FastEthernet0/23
--More--      switchport access vlan 230
--More--      switchport mode access
--More--      spanning-tree portfast
--More--      !
--More--      interface FastEthernet0/24
--More--      switchport access vlan 230
--More--      switchport mode access
--More--      spanning-tree portfast
--More--      !
```

```

--More--    interface GigabitEthernet0/1
--More--    switchport trunk allowed vlan 230
--More--    switchport mode trunk
--More--    !
--More--    interface GigabitEthernet0/2
--More--    !
--More--    interface Vlan1
--More--    no ip address
--More--    !
--More--    interface Vlan230
--More--    ip address 172.16.123.50 255.255.255.240
--More--    !
--More--    ip default-gateway 172.16.123.49
--More--    ip http server
--More--    ip http secure-server
--More--    !
--More--    line con 0
--More--    line vty 5 15
--More--    !
--More--    end

```

4.Evidencia de PING entre switches

```

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.123.66, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/4/9 ms
L3-0403#ping 172.16.123.50

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.123.50, timeout is 2 seconds:
..!!!
Success rate is 60 percent (3/5), round-trip min/avg/max = 1/1/1 ms
L3-0403#ping 172.16.123.50

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.123.50, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
L3-0403#
CTRL-A Z for help | 9600 8N1 | NOR | Minicom 2.11.1 | VT102 | Offline | ttyUSB0

```

L3 a L2 los 2

, de ser necesarias

lab-3-5@l

```
lab-3-5@lab-3-5-WS-13:~$ ping 172.16.123.66
PING 172.16.123.66 (172.16.123.66) 56(84) bytes of data.
 64 bytes from 172.16.123.66: icmp_seq=1 ttl=64 time=0.835 ms
 64 bytes from 172.16.123.66: icmp_seq=2 ttl=64 time=0.588 ms
 64 bytes from 172.16.123.66: icmp_seq=3 ttl=64 time=0.642 ms
 64 bytes from 172.16.123.66: icmp_seq=4 ttl=64 time=0.569 ms
 64 bytes from 172.16.123.66: icmp_seq=5 ttl=64 time=0.579 ms
 64 bytes from 172.16.123.66: icmp_seq=6 ttl=64 time=0.443 ms
 64 bytes from 172.16.123.66: icmp_seq=7 ttl=64 time=0.616 ms
 64 bytes from 172.16.123.66: icmp_seq=8 ttl=64 time=0.530 ms
 64 bytes from 172.16.123.66: icmp_seq=9 ttl=64 time=0.471 ms
 64 bytes from 172.16.123.66: icmp_seq=10 ttl=64 time=0.467 ms
 64 bytes from 172.16.123.66: icmp_seq=11 ttl=64 time=0.434 ms
 64 bytes from 172.16.123.66: icmp_seq=12 ttl=64 time=0.612 ms
 64 bytes from 172.16.123.66: icmp_seq=13 ttl=64 time=0.448 ms
 64 bytes from 172.16.123.66: icmp_seq=14 ttl=64 time=0.439 ms
 64 bytes from 172.16.123.66: icmp_seq=15 ttl=64 time=0.428 ms
 64 bytes from 172.16.123.66: icmp_seq=16 ttl=64 time=0.628 ms
 64 bytes from 172.16.123.66: icmp_seq=17 ttl=64 time=0.419 ms
 64 bytes from 172.16.123.66: icmp_seq=18 ttl=64 time=0.570 ms
 64 bytes from 172.16.123.66: icmp_seq=19 ttl=64 time=0.547 ms
 64 bytes from 172.16.123.66: icmp_seq=20 ttl=64 time=0.479 ms
 64 bytes from 172.16.123.66: icmp_seq=21 ttl=64 time=0.469 ms
 64 bytes from 172.16.123.66: icmp_seq=22 ttl=64 time=0.599 ms
 64 bytes from 172.16.123.66: icmp_seq=23 ttl=64 time=0.401 ms
 64 bytes from 172.16.123.66: icmp_seq=24 ttl=64 time=0.476 ms
^C
--- 172.16.123.66 ping statistics ---
24 packets transmitted, 24 received, 0% packet loss, time 23548ms
rtt min/avg/max/mdev = 0.401/0.528/0.835/0.098 ms
lab-3-5@lab-3-5-WS-13:~$
```

Ping L2 isla 3 a L2 isla 4

```
lab-3-5@lab-3-5-WS-14:~$ ping 172.16.123.50
PING 172.16.123.50 (172.16.123.50) 56(84) bytes of data.
From 172.16.123.65 icmp_seq=2 Redirect Host(New nexthop: 172.16.123.50)
From 172.16.123.65 icmp_seq=3 Redirect Host(New nexthop: 172.16.123.50)
From 172.16.123.65 icmp_seq=1 Destination Host Unreachable
From 172.16.123.65 icmp_seq=5 Redirect Host(New nexthop: 172.16.123.50)
From 172.16.123.65 icmp_seq=6 Redirect Host(New nexthop: 172.16.123.50)
From 172.16.123.65 icmp_seq=4 Destination Host Unreachable
From 172.16.123.65 icmp_seq=8 Redirect Host(New nexthop: 172.16.123.50)
^C
--- 172.16.123.50 ping statistics ---
8 packets transmitted, 0 received, +7 errors, 100% packet loss, time 7177ms
pipe 3
lab-3-5@lab-3-5-WS-14:~$
```

Ping L2 isla 4 a isla 3





